

Rehabix: A Real-Time AI-Based Posture Correction System for Physiotherapy and Rehabilitation

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ABSTRACT In this research paper, we develop a system that detects and corrects human posture in real-time while performing physiotherapy exercise routines. The developed system utilizes computer vision and machine learning methods to classify the correctness and incorrectness of a variety of exercises that include shoulder/arm lift exercises, hip/leg lift exercises, and squat exercises. Using video input, the human body will be recorded while performing selected exercises, and the selected exercises will have key skeletal landmark points extracted from a pose estimation framework. These extracted key points along with other extracted features will then be processed by a random forest classifier to determine the correct or incorrect classification of postural positioning. In addition, a real-time feedback loop has been developed into the system; therefore, you will be able to visually see your postural alignment immediately following your performance of an exercise. By utilizing this functionality, you will be able to modify your posture based on immediate visual feedback, thereby decreasing the chances of sustaining an injury during exercise. Based upon experimental evaluation of the developed model, we have shown that the developed posture classification achieves a high level of accuracy and reliability and the classification via the developed posture classification operates at an efficient level to enable the use of the developed model in a real-time fashion. As a result of these accomplishments, the developed system is a low-cost and easily accessible alternative to the traditional methods of providing physiotherapy supervision or using wearable devices to accomplish physiotherapy; therefore, the developed system will be suitable for use in rehabilitation, fitness, and home exercise guidance.

INDEX TERMS Computer vision, exercise correction, MediaPipe, physiotherapy, pose estimation, random forest, real-time feedback.

I. INTRODUCTION

GOOD Posture is an important component of both physiotherapy and exercise, and directly affects the quality of your exercise, injury prevention, rehabilitation results, and general musculoskeletal health. When you have good posture, you can achieve the best alignment from a biomechanical perspective, produce efficient muscle activation, distribute loads evenly between joints, and reduce loading strain during movement (hence producing less mechanical stress on your body). On the other hand, when you do not maintain good posture (or perform exercises incorrectly), you run the risk of developing overuse injuries as well as degeneration in the joints of the body and experiencing chronic pain or long-term musculoskeletal disorders (MSDs). The use of improper

movement patterns during physical activity and rehabilitation is one of the strongest indicators of injury and decreased effectiveness of treatment and progression through disorders such as LBPs, neck pain, shoulder dysfunction, and instability of the knee.

Globally, the number of people living with MSDs continues to increase at an alarming rate. The World Health Organisation (WHO) estimates that there are approximately 1.71 billion people worldwide currently living with a musculoskeletal condition, making MSDs the most significant cause of disability globally. LBP alone affects over 619 million people globally and will likely continue to increase to 843 million by 2050. Globally, LBP contributes almost 64 million years of disability (YLDs), while neck pain contributes ap-

proximately 22 million YLDs, and osteoarthritis contributes nearly 19 million YLDs. Research has indicated that incorrect posture and movement patterns while exercising can have a negative effect on rehabilitation progress, resulting in a decreased effectiveness of treatment (by more than 30% for those using incorrect movement patterns). Continuing to perform exercises incorrectly increases the chance of reinjury. In addition, sedentary behaviour has been shown to have a negative effect on postural health; with people being more physically inactive, by spending prolonged periods at desks or in front of screens, or not moving around enough, there is a significant increase in postural disorders for both young and older populations.

There has been a rapid escalation of the amount of physical inactivity experienced throughout the world. According to data from a recent worldwide study published in *The Lancet*, approximately 31.3% of adults worldwide = approximately 1.8 billion people) fail to meet the current recommendations for physical activity, an increase from 23.4% in 2000 = an increase of almost 5 percentage points in 20 years). This increase indicates that the worldwide trend in physical inactivity has continued to worsen over that period of time. More than 24.3 percent of adults in the United States are not meeting the recommended quantities of BOTH aerobic exercise and muscle strengthening exercise, with physical inactivity costing the U.S. economy an estimated \$117 billion in annual healthcare costs. Other studies indicate that the total economic cost from physical inactivity is approximately \$67.5 billion every year throughout the world = an estimate based on healthcare expenditures and productivity only).

In addition, there is a growing demand for rehabilitation services; the global rehabilitation market was approximately USD 7.62 billion in 2024 and is projected to exceed USD 12.4 billion by 2033 at a compound annual growth rate (CAGR) of about 6.3%. The digital rehabilitation and physiotherapy markets have seen tremendous growth due to the adoption of telehealth and increasing patient preference for home-based rehabilitation. The global home fitness and digital exercise support market is currently valued at over USD 11 billion. Since the onset of the COVID-19 pandemic, the number of people trying to exercise independently without guidance has greatly increased.

Despite the increasing emphasis on rehabilitation and exercise, physiotherapy services are still limited in many regions across the world. There is a critical shortage of rehabilitation professionals globally. In the United States there are over 260,000 physiotherapists servicing a population with an increasing need for rehabilitation, whereas many developing and underdeveloped countries do not have nearly enough rehabilitation providers to offer adequate amounts of rehabilitation care. For example, in India, the National Digital Health Programme has expanded access to healthcare for hundreds of millions of people, yet the ratio of physiotherapist to patients is inadequate to ensure consistent, supervised rehabilitation. Therefore, the combination of growing numbers of chronic conditions and an ageing population has created a compelling

case for scalable technology-supported rehabilitation.

Physiotherapy exercise is critical to enhancing mobility, improving function, and preventing decline; however, these exercises will not achieve their desired effects without the appropriate technique and consistency of execution. When patients are attempting exercises at home without immediate feedback regarding their performance, it is often difficult for them to maintain a correct posture, avoid compensatory movement patterns and achieve full ranges of motion while performing the exercise. As such, it is common for compliance to an unsupervised rehabilitation program to fall to less than 50%, and in many cases, the number of patients performing the exercises correctly while at home has been reported to exceed 70%. Therefore, there is a critical need for a system that is capable of monitoring posture and providing real-time corrective feedback to patients exercising outside of traditional clinical settings.

Recent advancements in artificial intelligence, computer vision and machine learning have provided new possibilities for addressing these problems. In particular, vision-based human pose estimation has been developed as a tool that can be used for intelligent rehabilitation monitoring. Unlike traditional motion capture systems, most of which require specialized wearable sensors, markers or other expensive technology to capture motion, most modern pose estimation systems extract body landmarks from standard RGB cameras; therefore pose analysis can occur using an inexpensive camera system.

Computer vision systems can analyze the joint alignment of the patient's body, determine if there are deviations from normal movement patterns and classify the exercise as having been performed correctly or incorrectly. Additionally, using computer vision to capture and analyze patient body movements in real-time allows patients to receive immediate feedback on their exercise performance.

Nevertheless, there are still many limitations to currently available technologies. Wearable sensor systems will often require hardware that may be uncomfortable for users. Although marker-based motion capture systems are very precise, they can be costly and impractical for everyday home use. Vision-based systems based on deep learning can also yield outstanding results but typically require extensive processing power which limits their ability to run in real-time on consumer devices. Many of these solutions focus mainly on classifying poses and do not use real-time corrective feedback tools to help support rehabilitation.

Motion monitoring has changed significantly over the years, from methods that rely on using markers attached to the subject to newer methods that utilize computer vision without the use of markers. The use of the Microsoft Kinect as part of a rehabilitation programme showed some promise; however, there were many issues related to lighting and occlusion. The use of light-weight models and single-camera systems has reduced the dependency on hardware in the development of new rehabilitation techniques. However, many of these techniques continue to focus on recognizing gestures instead

of providing a comprehensive corrective solution as part of the rehabilitation process.

This research presents an approach to using pose estimation along with a machine learning technique in order to provide real-time posture detection and correction through physiotherapy exercise monitoring.

This system will analyze common exercises that physical therapists prescribe including arm raises, leg raises and squats, which are common movements in rehabilitation, but people often do them incorrectly when they are not supervised. This system uses a combination of video inputs from a standard camera to provide an estimate of the position of skeletal points in space to evaluate whether someone is performing those movements correctly or how they are not and will give immediate feedback to users visually.

The proposed solution aims to combine the accuracy and accessibility of previous solutions. It will offer a system that is cost effective, easy to use, and can be deployed on commonly found consumer grade hardware that is also very computationally efficient. The goal of the system is to provide users with immediate correctional feedback while monitoring their exercises in order to help improve exercise accuracy and decrease the risk of injury, improve adherence to rehabilitation, and provide users with rehabilitation-quality therapy at home independent of supervision.

This project hopes to address the growing global challenge of the burden of musculoskeletal diseases and rehabilitation, with use of intelligent digital health technologies. Improvements to posture monitoring and to allow corrective feedback through cost-effective methods of using computer vision will help to make the delivery of physiotherapy support more accessible and decrease the wider burden created by individuals performing exercises incorrectly and not adequately supervised when rehabilitating from injuries.

A. PROBLEM STATEMENT

At present, there are no readily available hardware-free systems which are able to accurately classify multiple rehabilitative exercises simultaneously and provide immediate biomechanically valid corrective feedback while performing the exercises with primarily consumer grade devices and without cloud connectivity or wearables. Although mobile apps, videos and online fitness websites are becoming increasingly popular, most provide passive assistance and do not actively monitor and correct the users postural position while performing the exercise. Therefore, the effectiveness of these systems are dramatically reduced, especially when users rely on self-directed rehabilitation.

The existing systems are either using expensive depth sensing hardware (e.g. Kinect), limiting feedback to simple exercise recognition without correction, suffer from poor frame based predictions, or require cloud based processing which causes unacceptable latency for interactive rehabilitation. The systems using special hardware become costly and may require significant time to setup which limits their use to non-consumers. Cloud based solutions introduce concerns

for latency, privacy, and continual need for Internet access, further reducing the practical usability of these solutions for home use. Additionally, many existing solutions fail to provide consistent and temporally stable predictions, leading to unreliable feedback that can confuse users rather than assist them.

Individuals who exercise at home and rehabilitate themselves are not able to access any guidance about how to perform their exercises or use any corrective measures that would help them avoid making mistakes in their posture. By having the improper posture while doing physiotherapy exercises will reduce the value of that exercise, and may lead to increased problems with an existing condition (e.g., will likely cause them to re-injure themselves) or lead to the present a new injury. Therefore, there is a great need for an intelligent system that can guide exercise users accurately and consistently regardless of a professional's presence or absence to guide them as they do so.

Thus, this project's central issue is: How can a fully vision-based, locally-executed, and computationally simple solution to provide valid, real-time corrections of posture of physiotherapy users while they are exercising and without professional assistance, and do so using standard consumer hardware? This type of system must integrate the four elements of accuracy, efficiency, accessibility, and usability, so that the single system will perform all of the aforementioned functions in an integrated manner.

In this proposal, we propose a way to create a posture-of-the-user analysis and correction system that uses pose estimation and machine learning to assess the user as he/she performs a physiotherapy exercise in realtime. For this project, the proposed system will analyze a number of different types of common exercises that tend not to be performed correctly by the average, unsupervised physiotherapy user. Execution of these commonly done exercises include arm elevating, leg elevating, and squats and many times involve more than one joint and coordinated motion which make them good methods to evaluate both the accuracy of posture of the user and the effectiveness of the system to accomplish its intended purpose.

With the help of the MediaPipe Pose framework for skeletal keypoint extraction from video input and using a random forest classifier on the extracted keypoints to classify the correctness of the exercise, this works by identifying skeleton joint positions and using them to generate additional useful features such as joint angle measurements or relative distances between joints to improve the classification of correctness.

The skeletal measurement data can be processed into visual feedback to the user in real-time, allowing the user to correct their form/technique while still exercising. This increase in user awareness results in lower injury rates and improved exercise accuracy by reinforcing good posture throughout the entire duration of the exercise.

The system was created to provide efficient solutions for everyday users and can be implemented on any consumer-

level device; it does not require any specialized hardware and does not require continuing access to the internet, making it very easily accessible.

The goal of this solution is to be very accessible, inexpensive, and non-invasive for correcting posture and can be applied in multiple environments, including physical therapy, fitness training, and in-home rehabilitation, thereby addressing the significant global need for supervised exercise assistance that has been previously reported.

II. LITERATURE REVIEW

There has been a great deal of advancement in the area of assessing human posture and exercise over the last 20 years from original laboratory-based systems using markers to completely camera-based systems without markers. Most systems in the past used body-worn inertial measurement units (IMUs) or optical markers to measure the locations of joints, and therefore required controlled laboratory environments with specialists to monitor them, making them intrusive and costly, and not useful for use at home [16]. They also had limited capabilities for scalability and widespread use.

With the introduction of depth sensors used in cameras (e.g., Microsoft Kinect), there are now additional methods for markerless motion analysis. An early example of this is the work done by Vakanski, et al., developing a rehabilitation assistance monitoring system using Kinect sensors and modeling the timing of exercises using hidden Markov models (HMMs) [3]. Although they were effective at identifying different type of exercises in a controlled environment, they required specific Microsoft Kinect hardware which is no longer commercially available, and they worked poorly outdoors and in varying temperature environments. The same issues exist for the work performed by Chen, et al. using depth camera fusion methods combined with inertial sensors to identify actions in [4]. This method also requires tight hardware synchronization and adds to the complexity of the system, and thus it is not practical to use in large or cost-effective systems.

Prior to the advent of deep learning-based pose estimation, pose estimation had been heavily reliant on specialised hardware. OpenPose, developed by Cao et al., is a real-time multi-person pose estimation framework that uses Part Affinity Fields (PAFs) to incorporate and associate multiple body keypoints from standard RGB video accurately [1]. Although OpenPose has demonstrated high accuracy and supports multi-person tracking, the significant amount of computational resources required to run it — especially in real-time settings — prohibits deployment on resource-constrained devices unless the device has GPU acceleration.

In contrast, Google's MediaPipe Pose, a lightweight and efficient single-person pose estimation solution capable of identifying 33 three-dimensional body landmarks with low latency on CPU-only systems, has already demonstrated potential for use in real-time applications on consumer-grade devices such as laptops or smartphones [2]. However, to date, studies that have utilised MediaPipe for developing phys-

iotherapy specific posture correction systems with precise biomechanical validation and real-time corrective feedback have been limited primarily to using MediaPipe for gesture recognition and yoga pose classification.

The field of human pose estimation and the monitoring of intelligent exercise performance have evolved significantly over the past 20 years; from traditional marker based motion tracking systems, to vision-based systems using advanced machine learning algorithms to achieve rehabilitation. In the past, rehabilitation monitoring technologies used a combination of optical marker systems, inertial measurement units (IMUs) and wearable sensors to track joint movements and other movement characteristics. While these systems were able to provide accurate biomechanical data, the use of special hardware, the need for extensive calibration procedures and the necessity for controlled environments limited both the scalability and accessibility of these systems, particularly when used in rehabilitation settings outside the clinic.

The development of markerless computer vision methods has created a paradigm shift in the area of posture monitoring research. Depth cameras, particularly those developed on the Microsoft Kinect platform, were some of the first affordable, practical solutions to collect motion data for use in rehabilitation. The Kinect system was able to demonstrate that skeletal tracking can be achieved at a low cost and provides clinicians and researchers with practical means to assess the quality of a patient's postural alignment as they perform martial arts movements, conduct a gait analysis or provide corrective feedback. Studies have shown that posture classification accuracies exceed 85%-90% when Kinect has been used for monitoring postural alignment in controlled lab conditions. However, while the Kinect system has demonstrated strong potential for improving rehabilitation outcomes through precise monitoring, it suffers from limitations such as sensitivity to occlusion, reliance on specialized depth sensing hardware, poor outdoor performance, and variable tracking capabilities due to inconsistent lighting conditions. Furthermore, now that Kinect hardware has been discontinued, these methods have limited practicality as a means of enabling scalable deployment of a low-cost posture monitoring system.

In recent years, an increase in research focused on RGB camera-based pose estimation has occurred. Increasingly, researchers have addressed the hardware dependency of these methods using computer vision and machine learning. The main the methods use extracted skeletal landmarks and joint angle data in combination with biomechanical relationships to classify posture quality and detect deviations from normal movement.

Classical machine learning models have proven to be particularly effective when classifying posture due their efficiency, low complexity and interpretability, making them ideal for relatively small rehabilitation datasets. Random Forest is among the traditional machine learning classifiers that have become widely utilized for posture classification. An algorithm created by Breiman, Random Forest uses an ensemble of classifiers (i.e., decision trees) and has shown ex-

ceptional performance for classification problems due to the unique nature of the algorithm's ensemble structure, robust to noise and thus, has a weak tendency towards overfitting and the ability to model non-linear structure between skeletal features. For example, Liao et al. and others have shown that Random Forest tree models trained on joint-angle and distance based features consistently outperform single decision tree models and frequently achieve classification accuracies of 88–90%.

Similarly, Support Vector Machines (SVMs) have also been popular for use in posture recognition and rehabilitation monitoring. Due to their superior performance when used in high-dimensional feature spaces, SVMs are appropriate for classifying posture. SVM classification accuracies have been reported as being in the range of 85–87% across multiple datasets of different exercise modes; however, SVMs can be more complex to train than tree-based classifiers as they require careful kernel tuning.

Postural recognition and rehab monitoring have increasingly been recourse to support vector machines (SVMs), which have demonstrated successful high-dimension feature space performance for pose classification with 85% to 87% reported accuracy from a number of exercise datasets. However, SVMs generally require careful tuning of their kernel parameters and have a higher training complexity than tree-based models. Because they are easy to interpret and have low computational cost, logistic regression has been considered as a light-weight alternative to SVM; however, with the exception of some studies, most logistic regression studies produce lower accuracy results than SVMs, generally ranging from 80% to 83% for complex multi-class movement recognition tasks.

K-nearest neighbors (KNN) can also be useful for pose classification problems due to its simplicity of implementation and fast training time; however, the inference speed and scalability of KNN can create challenges for classification on large datasets, with KNN accuracy typically falling between 82% and 85%. Decision tree models are very computationally simple and understandable, but also display high levels of overfitting and lower overall generalization than many ensemble-based classifiers.

When viewed comparatively across all traditional classification models, the results illustrate a common theme regarding the trade-off among accuracy, cost of computation, and real-time suitability across all examined models. Random forest achieves a relatively high level of performance across classification accuracy, moderate training cost, fast inference time, and high robustness, thus making it a potentially ideal classifier to use for posture classification systems with a real-time response requirement.

Recent research has also increasingly explored deep learning approaches for posture and exercise recognition. Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) architectures have been used to capture spatial and temporal movement patterns directly from video sequences or skeletal tra-

jectories. CNN-based methods have demonstrated strong performance in extracting posture-related visual features, while LSTM models have been particularly effective for modeling temporal dependencies in sequential rehabilitation exercises.

Reported accuracies for deep learning-based exercise classification often exceed 90–95% under benchmark conditions, demonstrating their potential for sophisticated movement analysis. ResNet-inspired architectures and transformer-based pose models have further improved recognition accuracy for complex dynamic exercises. However, despite these advances, deep learning methods present several challenges in rehabilitation settings. They often require large labeled datasets, substantial computational resources, long training times, and hardware acceleration that may not be practical for consumer-grade rehabilitation systems. Additionally, labeled physiotherapy datasets remain relatively scarce compared to general human activity recognition datasets, increasing overfitting risks. To address limited training data, some studies have employed augmentation strategies such as scaling, rotation, translation, and synthetic noise injection to improve model robustness.

There is currently a growing body of literature indicating that incorporating rule-based biomechanical validation with ML-based approaches allows for improved accuracy when evaluating rehabilitation treatments. Hybrid frameworks have the potential to be an exciting new area in rehabilitation application development since they will use not only a data-driven classification, but will also have biomechanical or anatomical constraints to verify that movement is occurring accurately and safely. Mehrizi et al. (2016) demonstrated that when using anatomical joint angle thresholds (i.e., boundaries of normal motion within the body) in conjunction with machine learning classifiers, they were able to lower false positive rates and create clinically relevant interventions. In a similar study, Kim et al. (2014) presented a hybrid squat analysis system that was able to decrease the number of unsafe posture misclassifications by approximately 23.

Although the studies mentioned above indicate that progress is being made regarding hybrid approach technologies, there are still many exercise-specific applications that lack generalizability across all rehabilitation protocols. Additionally, there are no current examples of hybrid systems utilizing automated configuration of thresholds based on the type of exercise or user. Therefore, in an effort to broaden the application of hybrid approaches, the use of a modular rule engine utilizing configuration of exercise-specific thresholds presents an alternative method for providing individualized and flexible correction of posture.

The other major difficulty reported in the current literature is that there is temporal instability in frame-to-frame predictions/outputs. When classifying postures, most posture classification systems output predictions/classifications for each frame independently of one another. Therefore, individual frames may produce predictions which are not stable from one frame to another; thus resulting in frequent fluctuations in predicted classifications. The production of these kinds of

outputs can result in flickering feedback, decreased reliability of outputs, and reduced satisfaction of users while exercising continuously.

Wearable and Internet-of-Things (IoT) based rehabilitation systems have shown to provide continuously monitored data through the use of inertial sensors, electromyography and smart wearables. These systems provide accurate biomechanical measurements and have demonstrated their performance in monitoring patients. However, there are barriers to use these systems due to the costs of the devices, battery management, complexity of setup, and discomfort of use.

Vision based rehabilitation systems overcome many of these limitations by removing the need for any wearables. Recent lightweight pose estimation techniques have significantly reduced the computational demands for real-time inferences on consumer products. For instance, single camera pose estimation systems using landmark detection can achieve accurate posture estimation in real time at interactive frame rates without depending on cloud service providers.

Cloud based rehabilitation systems have also been created that leverage remote inference and telehealth integration. While these systems create more access points to digital rehabilitation, the dependence on the cloud can introduce latency in providing real-time feedback for corrective exercise. Some studies indicate that there is a delay associated with the network that exceeds one second, which can affect the ability to provide interactive exercise guidance. In contrast, locally deployed inference systems operating on standard consumer hardware can provide feedback in less than 0.5 seconds, making it suitable for real-time posture correction.

A review of the literature reveals that there are a number of important shortcomings within existing literature that have not been adequately addressed in prior research, including:

- Reliance on high-cost/discontinued hardware platforms (e.g., Kinect-based systems);
- Lack of exercise-specific feedback and guidance, with most approaches focusing on action recognition alone instead of providing rehabilitation guidance or corrective feedback;
- The inadequate capacity of purely data-driven systems to generalize across a range of lighting, camera placement, and body morphology;
- The instability in providing frame-level predictions without temporal smoothing or aggregation (i.e., without producing an average result of a specified number of predictions);
- The latency introduced when performing machine learning inference in the cloud; and
- The inability to provide biomechanically-safe feedback by integrating biomechanical constraints along with machine learning classifications, which may result in physiologically unsafe feedback.

To address these gaps, the proposed framework expands upon prior work while addressing its limitations. The framework consists of a modular, hybrid approach that combines: 1) machine learning-based posture classification; 2) configurable biomechanical validation rules; 3) temporal smoothing; and 4) local real-time inferences from a webcam. It

TABLE 1: Comparison of Machine Learning Models for Posture Classification

Model	Accuracy	Training Time	Prediction
Random Forest (Used model)	88–90%	Moderate	Fast
Support Vector Machine (SVM)	85–87%	High	Moderate
Logistic Regression	80–83%	Low	Very Fast
K-Nearest Neighbours (KNN)	82–85%	Very Low	Slow
Decision Tree	83–86%	Low	Fast

The comparison highlights the trade-offs between accuracy, computational complexity, and real-time performance. Random Forest achieves the best balance for posture classification tasks.

focuses on providing accessible, processor-efficient, and clinically meaningful feedback, and seeks to create an intelligent rehabilitation system that is applicable to both supervised and unsupervised settings.

III. PROPOSED WORK

This research proposes a new type of system that uses artificial intelligence (AI) to monitor posture and correct movement during the execution of physiotherapy or fitness exercises, so the exercises are performed correctly. The system employs computer vision, machine learning, and biomechanical rules as an analytical approach towards detecting human posture and giving immediate and actionable feedback to correct bad posture. The main objective is to bridge the gap between exercising under a physical therapist's supervision (i.e., in-clinic rehabilitation) versus exercising without supervision (i.e., home-based rehabilitation or fitness) by having an intelligent and automated guidance system.

The proposed system is entirely based on using a normal RGB camera, which is a significant difference compared to traditional systems that require the use of specialized sensors, need to be attached to the body with a wearable device, or require offline analysis. This has the effect of reducing the cost to set up and implement the system, as well as providing greater accessibility for many users. In fact, the system is designed to operate effectively with computers commonly available for consumers (i.e., desktop computers and laptops), allowing large numbers of users to adopt this new system in their home-based rehabilitation and fitness programs.

The primary design characteristics of the entire system are modularity, real-time responsiveness, interpretability, and accessibility for the user. Each element of the entire system is designed independently from one another; however, they are all integrated holistically. Therefore, each piece of the overall system will be easy to maintain and extend on an ongoing basis. Additionally, because the system is user-centric, all visual feedback and output of the proposed system will be visually intuitive and can be easily understood by non-technical users.

A. SYSTEM ARCHITECTURE

The system is built on a modular architecture composed of multiple layers where each distinct layer accounts for processing at biomechanical level. This architecture is advanta-

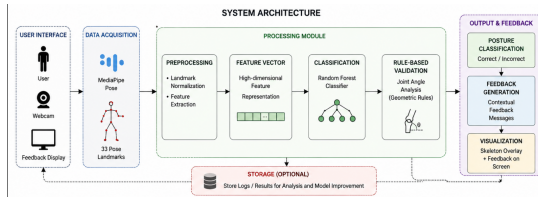


FIGURE 1. System architecture of the proposed posture correction system. The framework consists of multiple layers including input acquisition, pose estimation using MediaPipe, feature processing, machine learning-based classification, rule-based validation, and real-time feedback generation. The modular design ensures scalability, efficiency, and low-latency performance on consumer-grade hardware.

geous because it allows for modularity, thus allowing changes in one component to not affect all components of the entire system. It also allows for improvements and additions in the system, as well as efficient debugging.

1) Input Layer (Data Acquisition):

The input layer captures real-time video input from either a webcam or an embedded camera. The video output will be processed in continuous frames at approximately 24 to 30 (FPS), thereby providing smooth and continuous monitoring of user movement. A few requirements exist: a properly mounted and oriented camera (full view of the user's body), and sufficient ambient light to support effective detection. This layer initiates all processes that lead to the ultimate end user output.

2) Pose Estimation Layer:

The MediaPipe Pose framework processes video frames from the input layer and will perform real-time detection of 33 three-dimensional (3D) body landmarks located on various physical joints and interconnecting muscle groups of the user's body (e.g., shoulders, elbows, wrists, hips, knees, and ankles). The pose estimation model is also optimized to function efficiently and with very low latency on CPU-type systems without requiring GPU-type acceleration. This layer transforms raw visual input into structured skeletal representations, forming the foundation for further analysis.

3) Feature Processing Layer:

After skeletal landmarks have been isolated from the human body's video, they are turned into a structured numeric form for use by a computer. Each video frame results in a 99-dimensional feature vector (33 landmarks $\times$ 3 coordinates). Furthermore, raw coordinates are supplemented with higher-order biomechanical features such as joint angles/joint distance/relative body alignment. Higher-order features more accurately describe the dynamics involved with posture and movement than do raw coordinates only. Finally, normalizing of the feature values has taken place (typically via standard

scaling methods) to ensure compatibility between different subjects (due to different physical body composition) and different video camera calibration results.

4) Classification and Rule Engine:

The decision-making engine for the overall system is located here. The classification of postures as either correct or incorrect is performed via a Random Forest algorithm that uses extracted features or data from the Posture (3rd Layer). The Random Forest classifier was selected because of its robustness and high accuracy; furthermore, due to the Random Forest algorithm being an ensemble learning method (as well as the data-driven approach), it tends to be resistant against overfitting as compared to other classification algorithms. In conjunction with the Random Forest classifier is a rule-based validation module. This rule-based module examines the biomechanical constraints on the distance between joints, the range of joint angles, and the conditions under which two body parts are aligned relative to each other. By combining machine learning predictions and domain knowledge through the hybridization of a data-driven method and the use of biomechanical rules provides the highest level of reliability; the category of each posture remains physiologically relevant and safe after the rule engine has processed any potential overwriting or adjustments initiated through the machine-learning prediction.

5) Feedback and Visualization Layer:

The output will be displayed to users via an input screen (Visually with Visual Feedback and Textual Feedback). The system will be able to overlay skeleton keypoints on the video image stream so that users can visualize their posture in real time. The results of the posture classification will be indicated by a color coding scheme that is intuitively designed; i.e., Green = Good Posture; Red = Bad Posture. Additionally, users will have additional guidance for making specific corrections to their postures provided in the form of text; this is intended to increase user interest and make feedback immediately obvious and usable.

B. SYSTEM WORKFLOW

The system operates through a real-time sequential pipeline that transforms raw video input into actionable feedback:

- 1) Continuous capture of video frames from the webcam
- 2) Extraction of skeletal keypoints using pose estimation
- 3) Conversion of keypoints into feature vectors and joint angles
- 4) Classification of posture using a trained machine learning model
- 5) Validation using biomechanical rules
- 6) Generation of corrective feedback messages
- 7) Visualization of results with temporal smoothing for stability

A temporal smoothing mechanism (majority voting over recent frames) is applied to reduce prediction noise and ensure



FIGURE 2. Workflow of the real-time posture correction system. The system captures live video input, extracts skeletal keypoints, generates feature vectors, performs posture classification using a trained model, validates predictions through biomechanical rules, and provides real-time corrective feedback to the user.

consistent feedback.

C. METHODOLOGY

This methodology will involve following a structured and systematic approach that includes: data preparation, feature engineering, model training, and hybrid validation. These four main steps have been developed in a way that ensures each step has been developed with the utmost care and attention to detail so that they are robust, provide accurate results and are generalisable across different users and different environments.

1) Data Collection and Preprocessing

To collect appropriate data to develop the posture detection system a series of exercise videos representing both correct and incorrect postures were recorded for the following selected exercises; arm raises, leg raises and squat. These selected exercises have been identified as relevant to physiotherapy as well as having a greater chance of exhibiting posture related errors.

Each of the exercises was segmented into separate frames using OpenCV and collated into labelled datasets for each exercise based on whether the participant displayed the correct or incorrect posture while completing the exercise in question. A feature extraction algorithm called MediaPipe Pose was also applied to each of the frames in order to extract the skeletal keypoints from the exercises as well as dramatically reduce the overall dimensionality of the data while retaining the necessary and unique information required to represent the participants' postures. From a privacy perspective, this has also been achieved, by using the skeletal keypoints; it is not necessary to store any raw image data.

2) Feature Extraction

Each frame of the exercise will produce 33 landmarks (x, y, z coordinates) therefore producing a 99-dimensional feature vector. In order to increase classifier performance, additional biomechanical features will be computed for each joint; these will include; joint angle, limb orientation and position relative to the other joints of the human body. The additional

biomechanical features will enable the model to accurately differentiate between subtle variations of movement.

3) Data Augmentation

Data augmentation is used for improving the generalization of a model through the addition of augmented features to provide more evidence for the trained model, thus reducing any overfitting that may have occurred during training by enlarging the dataset. The augmentation techniques performed include the following:

1. Scaling of images or sample frames simulates the change in distance from the camera,
2. Translating images or sample frames to account for any shifts in position in an image or sample frame, and
3. Adding noise to each of the images or sample frames replicates the accuracy limitations of the sensor and changes in the environment.

When using these augmentations, we are able to increase the variability of the datasets while enhancing the ability of our trained model to perform well for different users, cameras, and the environments.

4) Model Training

A Random Forest classifier is developed for each exercise. Creating classifiers for distinct classes allows the model to create the posture specific to the exercise rather than using one generic model which can reduce accuracy on multiple exercise classifiers. The data set was properly divided between 80% training data and 20% testing data so that the model can have a proper evaluation of performance based on testing data.

Before the model can be trained, the features must first be normalized using the StandardScaler to ensure all features contribute equally to model and help create convergence. During classifier training, the model will be trained to identify two classes:

Proper Posture = 1

Improper Posture = 0

During classifiers training, the model will learn the patterns of the positions of joints, as well as angles between joints and distances between joints, which allow it to differentiate between proper execution of the exercise and improper execution of the exercise. Once the training process is completed, both the model and accompanying scaler must be saved and preserved for use during deployment and maintain consistency during real-time inference.

5) Hybrid Validation

The Biomechanical confirmation of the Human Movement Analysis model being developed is achieved by building a rule-based validation layer in addition to the current machine learning validation process. By combining the machine learning data obtained by numerous frames of video and rule-based validations, the system can continue to provide reliable and accurate predictions, based on the fundamentals of how the human body moves.

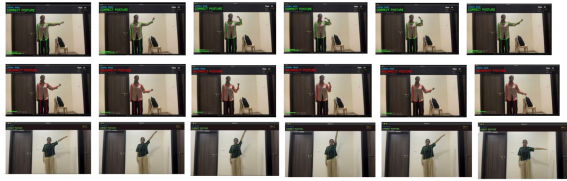


FIGURE 3. Real-time posture classification results for arm raise exercises. The figure presents multiple frames of correct (green) and incorrect (red) postures with skeletal overlays generated using pose estimation, demonstrating the system's capability to provide accurate and continuous feedback under varying conditions.

Validity rules for the arm raise, leg raise and squat, are built in accordance with principles of physiotherapy:

1. Arm raise should have elbow angles $\geq 160^\circ$. Wrist should be above shoulder.
2. Leg raise should have knee angles $\geq 170^\circ$.
3. Squat should maintain knee angles between 70° and 120° .

If there are discrepancies between the prediction and the biomechanical validation rules, the rule-based validation layer will either override or adjust the rule or biomechanical prediction, allowing for safe, interpretable feedback that is in accordance with current physiotherapy practice and safer than providing an incorrect prediction of movement analysis.

D. IMPLEMENTATION DETAILS

The system is implemented using Python-based open-source tools and is optimized for real-time performance.

1) Hardware and Software

Processor: Intel i5 or equivalent

RAM: 8 GB

Webcam: 720p/1080p

Libraries: MediaPipe, OpenCV, NumPy, Scikit-learn

The above listed requirements will allow the system to be accessible to the largest number of users as they will not be required to have expensive and/or custom-built systems.

2) System Modules

Frame Extraction Module

Pose Estimation Module

Data Augmentation Module

Model Training Module

Real-Time Feedback Module

Each of the above modules will operate independently of each other which will allow for modular development, ease of debugging and future extensibility. This type of pseudo-modular design will allow the developer to make updates to individual components without affecting the other parts of the system.

3) User Interface

The interface provides:

Real-time skeletal visualization

Posture classification (Correct/Incorrect)

Context-aware feedback messages

Session statistics (accuracy, duration, frame count)

With this approach, users should find it fairly simple to interpret the feedback given, making real-time corrections while performing the exercise.

E. SYSTEM PIPELINE AND ALGORITHM

The system operates in two phases:

1) Training Phase

Extract frames from videos

Generate pose-based feature vectors

Apply data augmentation

Train Random Forest model

Save model and scaler

2) Real-Time Inference Phase

Capture live video

Extract pose landmarks

Normalize features

Predict posture using ML model

Validate using geometric rules

Apply temporal smoothing

Display feedback

The integration of pose estimation, machine learning, rule-based validation, and smoothing techniques ensures accurate, stable, and real-time posture correction.

F. KEY DESIGN ADVANTAGES

The proposed system offers several advantages:

Real-Time Performance: Low latency enables immediate corrective feedback

Hybrid Validation: Combines ML and biomechanical rules for higher accuracy

Accessibility: Works with standard webcams, no specialized hardware required

Modularity: Easy to extend with new exercises or models

Scalability: Supports future integration of deep learning and cloud features

Reliability: Reduces false positives/negatives through validation and smoothing

IV. RESULTS AND DISCUSSION

The performance of the proposed real-time posture correction system was evaluated through both quantitative analysis and real-time experimentation. The results demonstrate the system's ability to accurately classify exercise postures while maintaining low latency and stable feedback under practical conditions.

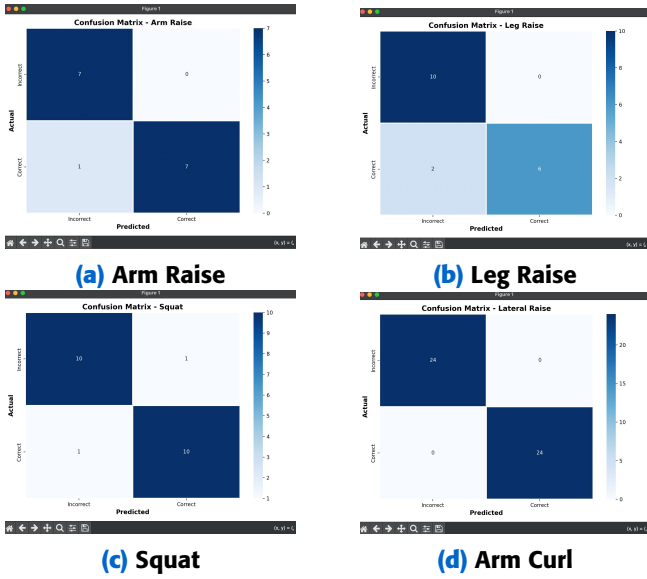


FIGURE 4. Confusion matrices for individual exercise models. Each matrix illustrates the classification performance for a specific exercise, showing true positives, true negatives, false positives, and false negatives. The results indicate high accuracy and balanced prediction capability across all exercise categories.

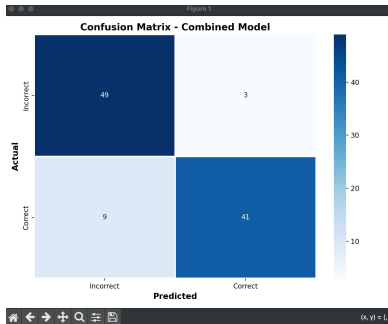


FIGURE 5. Confusion matrix of the combined posture classification model. The matrix demonstrates overall system performance across all exercises, highlighting high precision and recall with minimal misclassification.

A. QUANTITATIVE PERFORMANCE EVALUATION

The classification performance was assessed using exercise-specific datasets, with an 80–20 train-test split. The accuracy results for each exercise are summarized in Table 2.

The results indicate consistent performance across all exercise types. The slightly lower accuracy observed in squat classification is attributed to the increased complexity of multi-joint coordination and variability in squat depth among different users.

To further validate classification performance, confusion matrices for each exercise are presented in Fig. 2 (individual

TABLE 2: Model Performance Metrics for Posture Classification

Exercise	Accuracy
Arm Raise	97.75%
Leg Raise	100%
Squat	100%
Lateral Raise	98.6%

The table presents the classification accuracy achieved by the proposed system for different exercises. The results demonstrate consistent performance across all exercise types, with slightly lower accuracy observed in squat detection due to its higher biomechanical complexity.

models). These matrices illustrate true positives, true negatives, false positives, and false negatives, highlighting the model's balanced predictive capability across both correct and incorrect classes.

Additionally, the confusion matrix of the combined model (Fig. 3) demonstrates overall system performance across all exercises. The matrix shows high precision and recall, with minimal misclassification, confirming the robustness of the trained models.

B. REAL-TIME PERFORMANCE ANALYSIS

Beyond classification accuracy, the system was evaluated for real-time responsiveness, which is critical for interactive rehabilitation applications. The observed performance metrics are as follows:

Latency: 65 ms per frame

Frame Rate: 15–20 FPS

Feedback Delay: < 0.5 seconds

The integration of MediaPipe Pose with optimized processing through OpenCV enabled efficient landmark detection with minimal computational overhead. Despite operating on consumer-grade hardware, the system achieved smooth real-time performance.

The low latency ensures that users receive near-instantaneous feedback, allowing them to correct posture dynamically during exercise. Although the frame rate is slightly lower than high-end systems, it remains sufficient for accurate motion tracking and user interaction.

C. SYSTEM OUTPUT AND VISUALIZATION

The system provides intuitive visual and textual outputs to guide users effectively. Examples of real-time system output are shown in Fig 1.

These figures demonstrate how the system overlays skeletal landmarks on the live video feed while displaying posture classification (Correct/Incorrect) and contextual feedback messages.

The combination of skeletal visualization and corrective instructions enhances interpretability. Users can visually understand posture deviations while simultaneously receiving actionable guidance such as “Straighten your back” or “Raise your arm higher.”

D. COMPARATIVE MODEL ANALYSIS

Multiple machine learning models were evaluated, including Random Forest, Support Vector Machine (SVM), Logistic Regression, K-Nearest Neighbours (KNN), and Decision Trees.

Among these, the Random Forest classifier achieved the best overall performance. Its ensemble structure enables it to effectively capture non-linear relationships in high-dimensional pose data while maintaining robustness against noise and variations.

SVM provided competitive accuracy but required higher training time and computational resources.

Logistic Regression was computationally efficient but failed to model complex posture patterns.

KNN showed moderate accuracy but suffered from slow prediction speed, making it unsuitable for real-time use.

Decision Trees offered interpretability but were prone to overfitting.

These observations justify the selection of Random Forest as the primary model for the system.

E. DISCUSSION AND INTERPRETATION

The experimental results confirm that the proposed system achieves a strong balance between accuracy, efficiency, and usability. Several key observations can be drawn:

1) Accuracy and Reliability:

The system demonstrates stable classification performance across different exercises. The confusion matrices further validate that predictions are well-balanced, with low rates of misclassification.

2) Effectiveness of Hybrid Validation:

The integration of rule-based geometric validation with machine learning significantly improves reliability. This hybrid approach ensures biomechanical correctness, reducing both false positives and false negatives.

3) Real-Time Usability:

The system operates efficiently with low latency, enabling real-time feedback. Temporal smoothing further enhances usability by reducing flickering and stabilizing predictions.

4) User Experience:

Users reported that the feedback was clear, intuitive, and helpful in correcting posture. The combination of visual skeleton overlays and textual guidance improves learning and engagement.

5) Accessibility and Practical Deployment:

A key strength of the system is its ability to run on standard hardware using a webcam. This eliminates the need for expensive motion capture systems or wearable sensors, making the solution suitable for home-based rehabilitation.

F. LIMITATIONS

Despite strong performance, certain limitations remain:

- 1) The system currently supports a limited number of exercises
- 2) Performance may be affected under extreme lighting or occlusion conditions
- 3) The system primarily assumes a front-facing camera setup

These limitations highlight areas for future improvement.

V. CONCLUSION

This work presents a real-time posture correction system that integrates computer vision, machine learning, and biomechanical validation to deliver accurate and interpretable feedback for rehabilitation exercises. By leveraging pose estimation through MediaPipe and classification using a Random Forest model implemented via Scikit-learn, the system achieves high accuracy while maintaining computational efficiency.

The proposed solution successfully demonstrates that advanced posture analysis can be performed using only a standard webcam and open-source tools, without the need for specialized hardware. The ability to provide real-time, actionable feedback makes the system particularly valuable for physiotherapy, fitness training, and remote rehabilitation applications.

Furthermore, the hybrid validation approach ensures that feedback is not only data-driven but also grounded in biomechanical correctness, improving both safety and reliability. This combination of accuracy, interpretability, and affordability positions the system as a practical and scalable solution for real-world deployment.

A. FUTUTRE WORK

While the current system demonstrates strong performance, several enhancements can further extend its capabilities:

- **Expansion of Exercise Library:** Incorporating additional exercises (10–15 or more) to support a wider range of rehabilitation and fitness routines.
- **Integration of Advanced Models:** Implementing deep learning architectures such as Long Short-Term Memory (LSTM) networks and Transformer-based models to capture temporal dynamics of movement more effectively.
- **Mobile Application Development:** Developing smartphone-based applications to improve accessibility and portability, enabling users to perform exercises anywhere.
- **Progress Tracking and Analytics:** Adding features for long-term performance monitoring, including progress reports, trend analysis, and personalized recommendations for users and therapists.

APPENDIX A ALGORITHM DETAILS

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Algorithm 1 Real-Time Physiotherapy Posture Correction System

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1: PHASE 1: Training Pipeline
2: Input: Video dataset  $D = \{(v_i, y_i)\}$ , where  $y_i \in \{0, 1\}$ , exercise type  $E$ 
3: Output: Trained classifier  $RF_E$ , scaler  $SC_E$ 
4: for each exercise type  $E \in \{\text{ArmRaise, LegRaise, Squat}\}$  do
5:   Load video clips  $D_E$ 
6:   Initialize feature matrix  $X$ , label vector  $Y$ 
7:   for each video  $(v_i, y_i)$  in  $D_E$  do
8:     Extract frames  $F = \{f_1, f_2, \dots, f_n\}$ 
9:     for each frame  $f_k$  in  $F$  do
10:       $L_k \leftarrow \text{MediaPipe\_Pose}(f_k)$ 
11:      if  $L_k$  is None then
12:        continue
13:      end if
14:       $feat_k \leftarrow \text{Flatten}(L_k)$ 
15:       $aug \leftarrow \text{Augment}(feat_k)$ 
16:      Append  $aug$  to  $X$ ,  $y_i$  to  $Y$ 
17:    end for
18:  end for
19:  Split  $(X, Y)$  into train/test (80/20)
20:   $SC_E \leftarrow \text{StandardScaler.fit}(X_{train})$ 
21:  Normalize  $X_{train}$ 
22:   $RF_E \leftarrow \text{RandomForest.train}(X_{train}, Y_{train})$ 
23:  Save( $RF_E$ ,  $SC_E$ )
24: end for

25: PHASE 2: Real-Time Inference Pipeline
26: Input: Live webcam stream, exercise type  $E$ 
27: Output: Posture label and feedback
28: Load  $RF_E$ ,  $SC_E$ 
29: Initialize webcam  $CAM$ , buffer  $BUF$ 
30: while  $CAM$  is active do
31:   frame  $\leftarrow \text{capture}()$ 
32:   landmarks  $\leftarrow \text{MediaPipe\_Pose}(\text{frame})$ 
33:   if landmarks is None then
34:     continue
35:   end if
36:   feat  $\leftarrow SC_E.\text{transform}(\text{Flatten}(\text{landmarks}))$ 
37:   ml_label  $\leftarrow RF_E.\text{predict}(\text{feat})$ 
38:   angles  $\leftarrow \text{ComputeJointAngles}(\text{landmarks})$ 
39:   rule_label  $\leftarrow \text{RuleEngine}(\text{angles}, E)$ 
40:   if rule_label == Incorrect then
41:     final_label  $\leftarrow$  Incorrect
42:   else
43:     final_label  $\leftarrow$  ml_label
44:   end if
45:   Append final_label to  $BUF$ 
46:   if len( $BUF$ ) > window then
47:     Remove oldest element
48:   end if
49:   smoothed  $\leftarrow \text{MajorityVote}(BUF)$ 
50:   msg  $\leftarrow \text{GenerateFeedback}(\text{smoothed}, \text{angles}, E)$ 
51:   DrawSkeleton(frame)
52:   DisplayLabel(frame)
53:   DisplayMessage(frame, msg)
54:   Render(frame)
55: end while

```